Exercise 2

IAD2133 | OBJECT ORIENTED PROGRAMMING
Chapter 2 - Elementary Programming

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1. Scanner Class

Which method in the `Scanner` class is used specifically to read a value of type `double` from the console?

A. `nextDouble()`

B. `getDouble()`

C. `inputDouble()`

D. `readDouble()`

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2. Java Identifiers (True/False)

A Java identifier can legally begin with a digit, provided the rest of the characters are letters or underscores.

A. True

B. False

---

3. Valid Identifier

Which of the following is a valid Java identifier?

A. `123geeks`

B. `sum_&_difference`

C. `My Variable`

D. `$myvariable`

---

4. Modulus Operation

If you calculate `(6 + 10) % 7`, what is the result?

A. 1

B. 3

C. 2

D. 0

---

5. Floating-Point Literal

By default, a floating-point literal such as `5.0` is treated as which data type in Java?

A. float

B. long

C. double

D. int

---

6. Integer Division

What is the result of `5 / 2` in Java?

A. 2.5

B. 3

C. 2

D. 0

---

7. System Time

The method `System.currentTimeMillis()` returns the number of milliseconds elapsed since:

A. The time the program started

B. Midnight today

C. January 1, 1900 GMT

D. January 1, 1970 GMT

---

8. Type Casting

When casting a `double` to an `int`, such as `(int) 3.9`, what happens?

A. Fractional part is truncated

B. Stored elsewhere

C. Runtime error

D. Rounded

---

9. Software Development Phase

Which phase focuses on understanding the problem and documenting system requirements?

A. System Analysis

B. Requirements Specification

C. Implementation

D. System Design

---

10. Type Conversion

If one operand is `double` and the other is `int`, what is the result type?

A. float

B. long

C. int

D. double

---

11. System Analysis (True/False)

System Analysis involves identifying classes, interfaces, and relationships.

A. True

B. False

---

12. Augmented Assignment

Which is an example of an augmented assignment operator?

A. `+=`

B. `&&`

C. `++`

D. `==`

---

13. Acronym in System Design

What does IPO stand for?

A. Input, Process, Output

B. Application Programming Interface

C. Graphical User Interface

D. Integrated Development Environment

---

14. Expression Equivalence (True/False)

If `sum` is an integer, then `sum += 4.5` is equivalent to `sum = (int)(sum + 4.5)`.

A. True

B. False

---

15. Data Type Error

What error occurs when a value exceeds a data type's range?

A. Undeclared Variable

B. Integer Overflow

C. Round-off Error

D. Redundant Input

---

16. Maintenance Phase

Which phase focuses on improving the system and fixing bugs over time?

A. Implementation

B. Maintenance

C. Deployment

D. Testing

---

17. Expression Evaluation

What is the value of:

```
3 + 4 * 4 + 5 * (4 + 3) - 1
```

A. 83

B. 53

C. 112

D. 48

---

18. Identifier Length (True/False)

Identifiers in Java can be of any length.

A. True

B. False

---

19. Character Data Type

Which data type stores a single character in Java?

A. char

B. boolean

C. byte

D. String

---

20. Return Type (True/False)

`currentTimeMillis()` returns a value of type `int`.

A. True

B. False

Exercise code 1: Area of a Circle

Concept: Variables, Constants, and Basic Input.

Problem: Write a program that reads the radius of a circle from the user and computes the area. Use the constant Math.PI for the calculation.

- **Formula:** $Area = radius \times radius \times \pi$

Java

-
```java
import java.util.Scanner;

public class RadiusOfCircle {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.println("Enter the radius: ");
        double radius = input.nextDouble();

        double area = radius * radius * Math.PI;

        System.out.println("The area is = " + area);


    }

}
```

- OUTPUT:

-
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All ri

Install the latest PowerShell for new featu

PS C:\Users\ASUS> cd C:\OOP\
PS C:\OOP> java RadiusOfCircle
Enter the radius:
12
The area is = 452.3893421169302
PS C:\OOP>
```

Exercise code 2: Temperature Converter

Concept: Arithmetic Expressions and Floating-Point Math.

Problem: Write a program that reads a Celsius degree in a double value from the console, then converts it to Fahrenheit and displays the result.

- **Formula:** $Fahrenheit = (9 / 5) \times Celsius + 32$
- *Hint: Be careful with integer division when calculating 9/5!*

Java

```java
import java.util.Scanner;

public class TempreatureConverter {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.println("Enter the Celsius: ");
        double celsius = input.nextDouble();

        double farenheit = (9.0 / 5) * celsius + 32;

        System.out.println("Farenheit = " + farenheit);
    }
}
```

OUTPUT:

```
PS C:\OOP> java TempreatureConverter
Enter the Celsius:
15
Farenheit = 59.0
PS C:\OOP>
```

Exercise code 3: Time Decomposition

Concept: Integer Division (/) and Remainder Operator (%).

Problem: Write a program that prompts the user for a total number of seconds (e.g., 500) and converts it into minutes and remaining seconds (e.g., 8 minutes and 20 seconds).

Java

```java
import java.util.Scanner;

public class TimeDecomposition {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.println("Enter number of seconds: ");
        int seconds = input.nextInt();

        int minutes = seconds / 60;
        int remaining = seconds % 60;

        System.out.println(minutes + " minutes and " + remaining + " seconds ");

    }
}
```

OUTPUT:

```
PS C:\OOP> cd C:\OOP\
PS C:\OOP> java TimeDecomposition
Enter number of seconds:
30
0 minutes and 30 seconds
```

Exercise code 4: Digit Summation

Concept: Modulo and Division for digit extraction.

Problem: Write a program that reads an integer between 0 and 1000 and adds all the digits in the integer. For example, if an integer is 932, the sum of all its digits is $9 + 3 + 2 = 14$.

Java

```java
import java.util.Scanner;

public class DigitSummation {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.println("Enter number: ");
        int num = input.nextInt();

         int sum = 0;
        while (num > 0) {
            sum += num % 10;
            num /= 10;
        }

        System.out.println("Sum = " + sum);
    }
}
```

OUTPUT:

```
PS C:\OOP> java DigitSummation
Enter number:
932
Sum = 14
```

Exercise code 5: Gratuity Calculator

Concept: Combining multiple inputs and percentages.

Problem: Write a program that reads the subtotal and the gratuity rate (percentage), then computes the gratuity and the total. For example, if the user enters 10 for subtotal and 15% for gratuity rate, the program displays $1.5 as gratuity and $11.5 as total.

Java

```java
import java.util.Scanner;

public class GratuityCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter subtotal: ");
        double subtotal = input.nextDouble();

        System.out.print("Enter gratuity rate (%): ");
        double rate = input.nextDouble();

        double gratuity = subtotal * rate / 100;
        double total = subtotal + gratuity;

        System.out.println("Gratuity = " + gratuity);
        System.out.println("Total = " + total);
    }
}
```

OUTPUT:

```
PS C:\OOP> java GratuityCalculator
Enter subtotal: 10
Enter gratuity rate (%): 15
Gratuity = 1.5
Total = 11.5
PS C:\OOP>
```

Exercise code 6: Exponentiation and Math Functions

Concept: Using the Math.pow(a, b) method.

Problem: Write a program that displays a table of powers. For each number n from 1 to 5, calculate n^{n+1}.

- *Expected output row 1:* 1 2 1 (where 1 is n, 2 is $n+1$, and 1 is 1^2)

Java

```java
import java.util.Scanner;

public class PowerTable {
    public static void main(String[] args){
     for (int n = 1; n <= 5; n++) {
            System.out.println(n + " " + (n + 1) + " " + Math.pow(n, n + 1));
        }

    }
}
```

OUTPUT:

```
1 2 1.0
2 3 8.0
3 4 81.0
4 5 1024.0
5 6 15625.0
[Finished in 2.9s]
```

Exercise code 7: Numeric Type Casting

Concept: Implicit vs. Explicit Casting.

Problem: Create a program that prompts the user for three integers, calculates their average, and displays it as a double. Note: Ensure the result has decimal precision (e.g., the average of 1, 2, and 4 should be 2.333, not 2.0).

Java

```java
import java.util.Scanner;

public class Average {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter 3 integers: ");
        int a = input.nextInt();
        int b = input.nextInt();
        int c = input.nextInt();

        double avg = (a + b + c) / 3.0;

        System.out.println("Average = " + avg);
    }
}
```

OUTPUT:

```
PS C:\OOP> java Average
Enter 3 integers: 1
2
4
Average = 2.3333333333333335
PS C:\OOP>
```